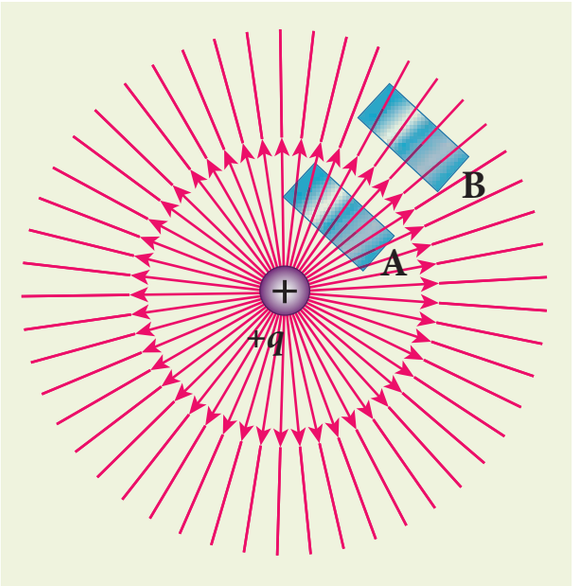


Figure 1.30 Electric flux

The electric field of a point charge is drawn in this figure. Consider two small rectangular area elements placed normal to the field at regions A and B. Even though these elements have the same area, the number of electric field lines crossing the element in region A is more than that crossing the element in region B. Therfore the electric flux in region A is more than that in region B. Since electric field strength for a point charge decreases as the distance increases, electric flux also decreases as the distance increases. The above discussion gives a qualitative idea of electric flux. However a precise definition of electric flux is needed.

Electric flux for uniform Electric field

Consider a uniform electric field in a region of space. Let us choose an area A normal to the electric field lines as shown in Figure 1.31 (a). The electric flux for this case is

$$\Phi_E = EA \qquad (1.52)$$

Suppose the same area A is kept parallel to the uniform electric field, then no electric field lines pass through the area A , as shown in Figure 1.31(b). The electric flux for this case is zero.

$$\Phi_E = 0 \qquad (1.53)$$

If the area is inclined at an angle θ with the field, then the component of the electric field perpendicular to the area alone contributes to the electric flux. The electric field component parallel to the surface area will not contribute to the electric flux. This is shown in Figure 1.31 (c). For this case, the electric flux

$$\Phi_E = (E\cos\theta)\, A \qquad (1.54)$$

Further, θ is also the angle between the electric field and the direction normal to the area. Hence in general, for uniform electric field, the electric flux is defined as

$$\Phi_E = \vec{E}\cdot\vec{A} = EA\cos\theta \qquad (1.55)$$

Here, note that $\vec{A}$ is the area vector $\vec{A} = A\hat{n}$. Its magnitude is simply the area A and its direction is along the unit vector $\hat{n}$ perpendicular to the area as shown in Figure 1.31. Using this definition for flux, $\Phi_E = \vec{E}\cdot\vec{A}$, equations (1.53) and (1.54) can be obtained as special cases.

In Figure 1.31 (a), $\theta = 0^o$. Therefore,

$$\Phi_E = \vec{E}\cdot\vec{A} = EA$$

In Figure 1.31 (b), $\theta = 90^o$. Therefore,

$$\Phi_E = \vec{E}\cdot\vec{A} = 0$$

UNIT-1(XII-Physics_Vol-1).indd 37 14-02-2020 19:33:09